Exercise 2.3

Calculate the probability that a life aged 0 will die between ages 19 and 36, given the survival function

$$S_0(x) = \frac{1}{10}\sqrt{100 - x}, \qquad 0 \le x \le 100 \ (= \omega).$$

Solution

The required probability is the probability that a life aged 0 survives to age 19 and then dies before age 36. In actuarial notation, this is

$$_{19|17}q_0 = {}_{19}p_0 \, {}_{17}q_{19}.$$

Equivalently, since the life is aged 0 and $S_0(0) = 1$, we can compute it using the survival function directly:

$$_{19|17}q_0 = S_0(19) - S_0(36).$$

Now,

$$\begin{aligned} S_0(19) &= \frac{1}{10}\sqrt{100 - 19} = \frac{1}{10}\sqrt{81} = \frac{9}{10}, \\ S_0(36) &= \frac{1}{10}\sqrt{100 - 36} = \frac{1}{10}\sqrt{64} = \frac{8}{10}. \end{aligned}$$

Therefore,

$$\begin{aligned} _{19|17}q_0 &= S_0(19) - S_0(36) \\ &= \frac{9}{10} - \frac{8}{10} \\ &= \frac{1}{10}. \end{aligned}$$

Thus, the probability that a life aged 0 will die between ages 19 and 36 is

$$\boxed{_{19|17}q_0 = \frac{1}{10} = 0.1}.$$